

Thibaut Cuvelier

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Senior software engineer (algorithms, optimisation), PhD-level background in reinforcement learning and high-performance C++.

Work experience, research projects, and education

Senior software engineer — Google Research (Operations Research) — Switzerland — since 2021-09

- Solving high-priority issues in LLM-based agents easing the use of our logistics APIs: evaluation (mixing formal methods and LLM-as-a-judge), code generation.
- Reduced logistics decision time from hours to seconds with millions of shipments (through online decision-making) and cut customer costs by approx. 30% by designing and implementing a distributed backend with distributed storage and graph algorithms in high-performance C++ that exploit problem-specific structure.
- Increased throughput of large-scale supply-chain simulations by 30x by adding features to an existing C++ solver for vector bin packing based on column generation and Gurobi.
- Mentored student researchers (machine learning) and a 20% collaborator, with accepted publications at ICML, NeurIPS, and JuliaCon.

Doctoral researcher / PhD student (computer science / mathematics) — Orange Labs (MORE) & université Paris-Saclay (CentraleSupélec, L2S) — France — 2017-10 to 2021-06

- Designed efficient algorithms for reinforcement learning (combinatorial bandits) with best-in-class runtime performance (polynomial) and statistical efficiency (logarithmic scaling). Performed a thorough theoretical analysis of the impact of the underlying approximation algorithm as well as numerical experiments.
- Slashed labelling efforts by up to 20% by developing a novel active-learning algorithm based on reinforcement learning while keeping or improving the target accuracy (sequential decision-making under uncertainty).
- Compared network routing under uncertainty and with fairness objectives. Solvers: Gurobi, Xpress, CPLEX, Mosek.

Research engineer — université de Liège — Belgium — 2015-09 to 2017-09

- Reduced water-depth safety margins by up to 50% in dam management, as measured by historical scenario simulations, by building and deploying a stochastic optimisation model under uncertainty in collaboration with domain experts. LP solvers: Gurobi, CPLEX.
- Reduced operating costs of industrial processes by up to 24% through a mathematical model for optimisation under uncertainty. LP solvers: Gurobi, CPLEX.

Intern — N-SIDE — Belgium — 2014-07 to 2014-08

- Improved existing software to optimise energy costs through robust optimisation in industrial plants within the existing Scala code base. Solvers: lp_solve, CPLEX.

MSc & BSc in computer engineering — université de Liège — Belgium — 2010-09 to 2015-06

Contributions to open-source projects: LyX, Qt, OR-Tools (C++), JuMP.jl, MathOptInterface.jl (Julia).

Publications to first-tier conferences: ICML, NeurIPS, ALT, ACM SIGMETRICS, IEEE TNSM.

Skills

Reinforcement learning (bandits), convex and combinatorial optimisation, algorithms and data structures.

Machine learning: scikit-learn, Pandas, SQL.

Programming languages: high-performance C++ (profiling, low-level optimisation), Python, Julia.